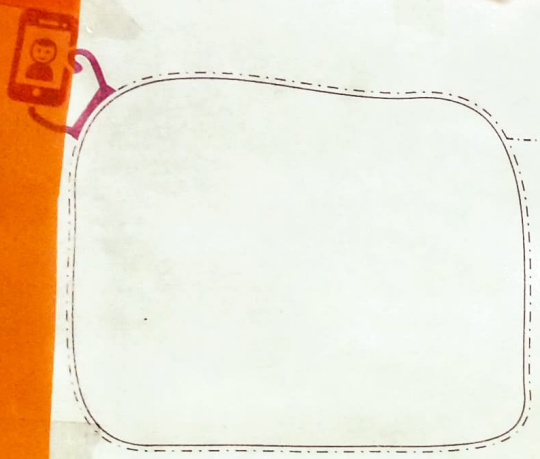


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Subject Anatomy SOL

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Cardiovascular System (CVS)

Introduction

The cardiovascular system is a closed circulatory system consisting of the heart, blood vessels and blood. It transports oxygen, nutrients, hormones and metabolic waste products throughout the body and helps maintain homeostasis.

Components of the Cardiovascular system

1. Heart: A muscular organ that pumps blood.
2. Arteries: Carry blood away from the heart.
3. Veins: Carry blood towards the heart.
4. Capillaries: Microscopic vessels where exchange of gases, nutrients and wa
5. Blood: A fluid connective tissue that transports substances throughout bo

The Heart

The heart is a hollow, muscular organ situated in the middle mediastinum of the thoracic cavity, between the two lungs. Approximately two-thirds of it lies to the left of the median plane.

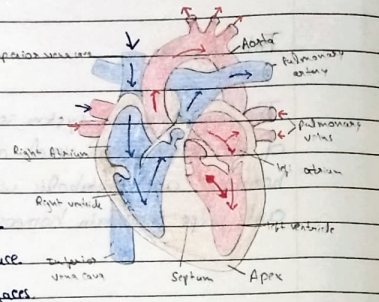
- Shape: Conical
- Apex: Directed forwards and to the left; normally located in th left fifth intercostal space near the midclavicular line.

- Base: Directed backwards and mainly formed by the left atrium
- Covering: Pericardium

External Features

The heart has:

- Apex and base
- Sternocostal (anterior) surface
- Diaphragmatic (inferior) surface
- Right and left pulmonary surfaces
- Coronary and interventricular grooves



Chambers of the heart

The heart consists of four chambers.

1. Right atrium: Receives deoxygenated blood from the superior vena cava, inferior vena cava and coronary sinus.

2. Right ventricle: Pumps deoxygenated blood into the pulmonary trunk and pulmonary arteries

3. Left atrium: Receives oxygenated blood from the four pulmonary vein

4. Left ventricle: Pumps oxygenated blood into the aorta for distribution throughout the body. Its wall is the thickest among the four chambers

Valves of the heart

The valves ensure unidirectional blood flow.

Valve	Location
Tricuspid valve	Right atrium and right ventricle
Pulmonary valve	Right ventricle and pulmonary trunk
Mitral (bicuspid) valve	left atrium and left ventricle
Aortic valve	left ventricle and aorta

Blood circulation

The cardiovascular system has two main circuits.

1. Pulmonary circulation: Deoxygenated blood travels from the right ventricle to the lung through the pulmonary arteries. Oxygenated blood returns to the left atrium through the pulmonary veins.

2. Systemic circulation: Oxygenated blood travels from the left ventricle through the aorta to the body. Deoxygenated blood returns to the right atrium through the vena cava.

Conducting system of the heart

The conducting system initiates and coordinates cardiac contraction

- Sinoatrial (SA) node: The normal pacemaker of the heart.
- Atrioventricular (AV) node: Delays the impulse into the ventricles. Activation
- AV bundle (bundle of His): Conducts impulses into the interventricular septum
- Right and left bundle branches: Carry impulses towards the ventricles
- Purkinje fibres: Distribute impulses through the ventricular myocardium.

Blood supply of the heart

The heart is supplied by the right and left coronary arteries, which arise from the ascending aorta.

- Right coronary artery (RCA): Commonly supplies much of the right side of the heart and parts of the conducting system.
- Left coronary artery (LCA): Divides mainly into the anterior interventricular artery (LAD) and circumflex artery.
- Venous drainage: Mainly through the coronary sinus into the right atrium.

Functions of cardiovascular system

1. Transport of oxygen and nutrients.
2. Removal of carbon dioxide and metabolic wastes.
3. Transport of hormones
4. Regulation of body temperature.
5. Maintenance of tissue perfusion and homeostasis
6. Support of immune defence and haemostasis

Clinical anatomy

- Coronary artery disease: Narrowing of coronary arteries may cause myocardial ischaemia or myocardial infarction.
- Valvular heart disease: Narrowing or incompetence of a valve may impair blood flow.
- Conduction abnormalities: Disturbances in impulse generation or conduction may produce arrhythmias.

Body tissues → Venae cavae → Right atrium → Right ventricle → Pulmonary arteries → lungs - gas exchange → pulmonary veins → left atrium → left ventricle → Aorta → Body tissue.

Conclusion: The cardiovascular system maintains circulation and tissue perfusion and its essential for the normal functioning of all organs.

Central Nervous System (CNS)

Introduction

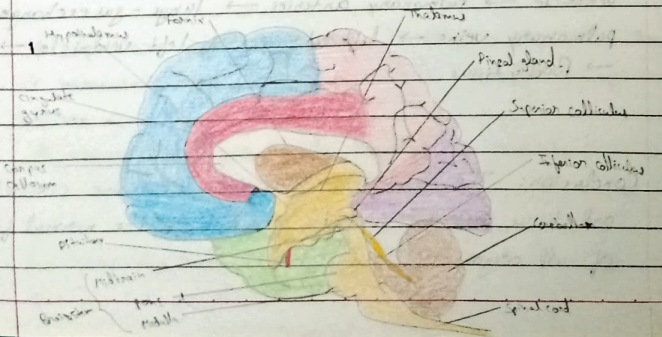
The central nervous system consists of the brain and spinal cord. It receives sensory information, integrates it and coordinates motor responses and higher mental functions.

Components of the CNS

1. Brain: Located in the cranial cavity and protected by the skull
2. Spinal cord: Located within the vertebral canal and continuous with the medulla oblongata

Brain

The brain is divided into the following major parts.



1. Cerebrum

The largest part of the brain, consisting of right and left cerebral hemispheres connected by the corpus callosum

The cerebral cortex contains grey matter, while deeper regions contain white matter and groups of grey matter called nuclei.

The cerebral hemisphere have four principal lobes:

- Frontal lobe: Voluntary motor activity, planning, behaviour and speech production
- Parietal lobe: Somatic sensation and spatial awareness
- Temporal lobe: Hearing, memory and aspects of language comprehension
- Occipital lobe: Vision

2. Diencephalon

It includes the thalamus and hypothalamus

- Thalamus: Major relay centre for sensory information reaching the cerebral cortex, with important exceptions such as olfaction
- Hypothalamus: Regulates autonomic functions, temperature, hunger, thirst and endocrine activity through its relationship with the pituitary gland.

3. Brainstem

The brainstem consists of:

- **Midbrain:** Contains important ascending and descending pathways and nuclei involved in eye movements and reflexes
- **Pons:** Connects parts of the brain and contains pathways and nuclei involved in the respiration and cranial nerve functions
- **Medulla oblongata:** Contains vital centres involved in respiration, cardiovascular regulation and protective reflexes such as swallowing

4. Cerebellum

The cerebellum lies in the posterior cranial fossa, behind the pons and medulla.

Functions:

- Coordination of voluntary movements
- Maintenance of posture and balance
- Regulation of muscle tone
- Motor learning

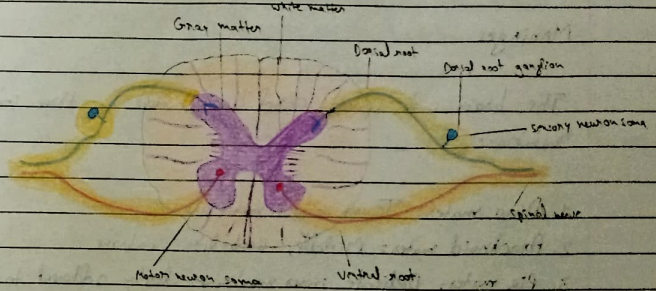
Spinal cord

The spinal cord is a cylindrical part of the CNS that extends from the medulla oblongata through the foramen magnum into the vertebral canal. In adults, it usually ends at the level of the L1-L2 vertebrae as the conus medullaris.

External features

- Cervical enlargement
- Lumbosacral enlargement
- Conus medullaris
- Cauda equina
- Filum terminale

The spinal cord gives rise to 31 pairs of spinal nerves.



Cross section of Spinal Cord.

Internal Structure

In transverse section:

- Gray matter: Central, H-shaped region containing neuronal cell bodies
- White matter: Peripheral region containing ascending and descending nerve fibres.
- Central canal: Small canal containing cerebrospinal fluid
- Posterior horns: Mainly sensory processing.
- Anterior horns: Contain lower motor neurons
- Lateral horns: Present primarily in certain spinal cord segments and contain autonomic neurons.

Meninges

The brain and spinal cord are covered by three connective tissue membranes.

1. Dura mater: Tough outer membrane
2. Arachnoid mater: Middle, web-like membrane
3. Pia mater: Delicate inner membrane closely adherent to the nervous tissue

The subarachnoid space, between the arachnoid and pia matter, contains cerebrospinal fluid (CSF)

Cerebrospinal fluid (CSF)

CSF is produced mainly by the choroid plexus of the ventricles.

Functions:

- Provides mechanical protection to the CNS
- Provides buoyancy to the brain.
- Helps maintain a stable chemical environment.
- Participates in the movement of nutrients and waste products.

Blood supply of the brain

The brain receives arterial blood through two paired systems:

- Internal carotid arteries: Supply much of the anterior circulation
- Vertebral arteries: Unite to form the basilar artery and contribute to the posterior circulation.

These systems communicate through the circle of willis at the base of the brain.

Functions of the CNS

1. Reception and integration of sensory information
2. Control of voluntary movement
3. Regulation of autonomic functions
4. Higher functions, including memory, learning, language and emotion.
5. Coordination of reflex responses and maintenance of homeostasis

Clinical anatomy

- **Stroke:** Brain injury caused by interruption of blood supply or haemorrhage.
- **Meningitis:** Inflammation of the meninges, commonly due to infection.
- **Spinal cord injury:** May cause sensory, motor and autonomic deficits below the level of injury.
- **Hydrocephalus:** Abnormal accumulation of CSF with the ventricular system.

Conclusion:

The CNS acts as the main integrating and controlling system of the body and coordinates sensory, motor, autonomic and higher mental functions.

Ulnar Nerve

Introduction

The ulnar nerve is a major terminal branch of the medial cord of the brachial plexus. It is primarily derived from C8 and T_1 spinal nerve roots.

It is a mixed nerve containing motor, sensory and autonomic fibers.

Root value

C_8, T_1 - with occasional contribution from C7.

Origin

The ulnar nerve arises from the medial cord of the brachial plexus, which is a continuation of the anterior divisions of the lower trunk.

Course and relations

1. In the arm

- Begins in the axilla as a branch of the medial cord.

- Descends along the medial side of the arm.
- Initially lies medial to the brachial artery.
- Pierce the medial intermuscular septum in the middle of the arm.
- Passes posterior to the medial epicondyle of the humerus in groove for ulnar nerve

2. At the elbow

The ulnar nerve passes behind the medial epicondyle, where it is relatively superficial and vulnerable to injury.

It then enters the forearm between the humeral and ulnar heads of the flexor carpi ulnaris.

3. In the forearm

- Descends deep to the flexor carpi ulnaris.
- Lies on the flexor digitorum profundus.
- In the distal forearm, accompanies the ulnar artery, generally on its medial side.
- Gives off the dorsal and palmar cutaneous branches

4. At the wrist and hand

The ulnar nerve enters the hand superficial to the flexor retinaculum through Guyon's canal, alongside the ulnar artery.

It divides into:

- Superficial branch
- Deep branch.

Branches of the ulnar nerve

Branch	Supply
Muscular branches	Flexor carpi ulnaris and medial half of flexor digitorum profundus.
Palmar cutaneous branches	Skin over the medial part of the palm
Dorsal cutaneous branch	Skin over the medial dorsum of the hand and medial digits
Superficial terminal branch	Palmaris brevis and digital sensory supply
Deep terminal branch	Most intrinsic muscles of the hand

Palmar cutaneous nerves



Muscles supplied

In the forearm

1. Flexor carpi ulnaris
2. Medial half of flexor digitorum profundus, supplying the ring and little fingers.

In the hand

The ulnar nerve supplies most intrinsic hand muscles, including:

- Hypothenar muscles
- Palmar and dorsal interossei
- Third and fourth lumbricals.
- Adductor pollicis
- Deep head of flexor pollicis brevis
- Palmaris brevis.

The thenar muscles are mainly supplied by the median nerve, with the deep head of flexor pollicis brevis being an important exception.

Sensory supply

The ulnar nerve provides sensation to:

- The little fingers.
- The medial half of the ring finger.
- The corresponding medial part of the palm and dorsum of the hand

Its palmar and dorsal cutaneous branches supply different areas of skin

Functions

1. Flexion and adduction of the wrist through flexor carpi ulnaris
2. Flexion of the distal interphalangeal joint of the ring and little fingers.
3. Abduction and adduction of the fingers.
4. Fine movements of the fingers and thumb
5. Sensation over the medial side of the hand

Clinical anatomy

1. Ulnar nerve injury at the elbow

Injury may occur behind the medial epicondyle.

Features include:

- Weakness of intrinsic hand muscles
- Difficulty abducting and adducting the fingers
- Sensory loss over the little finger and medial half of the ring finger.
- Ulnar clawing, particularly affecting the ring and little fingers

2. Froment's sign

When asked to hold a piece of paper between the thumb and index finger, a patient with ulnar nerve palsy may flex the terminal interphalangeal joint to compensate for weak adductor pollicis

3. Guyon's canal syndrome

Compression of the ulnar nerve at the carpi can cause weakness of intrinsic hand muscles and sensory changes in the palmar ulnar distribution. Dorsal ulnar hand sensation may be spared because the dorsal cutaneous branch usually arises proximal to the canal.

Conclusion:

The ulnar is essential for the fine motor movements of the hand and sensation over its medial aspect. Injury may significantly impair grip and finger coordination.

Brachial Plexus

Introduction

The brachial plexus is a network of nerves formed by the anterior (ventral) rami of spinal nerves C5, C6, C7, C8 and T1. It supplies the upper limb with motor, sensory and autonomic fibres.

Formation

The brachial plexus is formed by five roots, which unite to form three trunks. Each trunk divides into anterior and posterior divisions which regroup into three cords and finally give rise to terminal branches.

The sequence is:

Root → Trunks → Divisions → Cords → Branches

Parts of the brachial plexus

1. Roots

- The roots are the anterior rami of: C5, C6, C7, C8, T1
- They emerge between the anterior and middle scalene muscles in the neck.

⇒ Important branches arising from the roots:

- Dorsal scapular nerve - mainly C5
- long thoracic nerve - C5, C6 and C7

2. Trunks

The five roots unite to form three trunks

- Upper trunk - C5 + C6
- Middle trunk - Continuation of C7
- lower trunk - C8 + T_1

Important branches from the upper trunk include the suprascapular nerve and nerve to subclavius.

3. Divisions

Each trunk divides into:

- Anterior division.
- Posterior division.

Thus, there are six divisions in total.

Generally, the anterior divisions supply flexor compartments, while the posterior divisions supply extensor compartment of the limbs. This is a useful broad pattern rather than an absolute rule for every muscle.

4. Cords

The cords are named according to their relationship to the second part of the axillary artery.

- Lateral cord - Anterior divisions of upper and middle trunks
- Medial cord - Anterior division of lower trunk
- Posterior cord - Union of all three posterior divisions

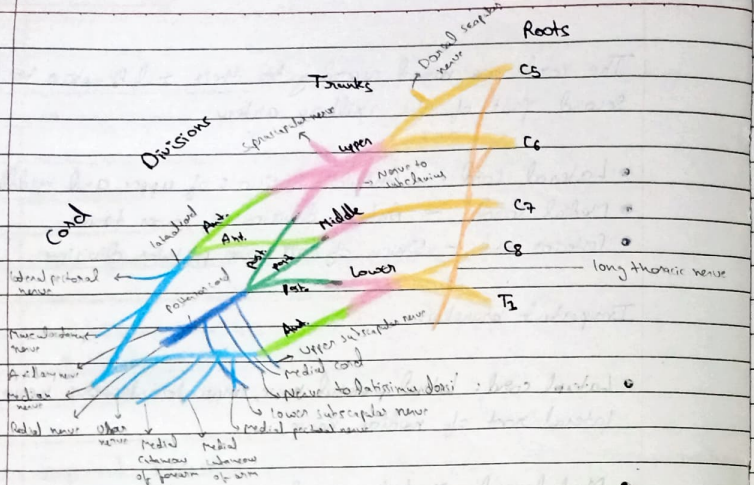
Important branches:

- Lateral cord: lateral pectoral nerve, musculocutaneous nerve and lateral root of median nerve.
- Medial cord: Medial pectoral nerve, medial cutaneous nerves of arm and forearm, ulnar nerve and medial root of median nerve.
- Posterior cord: Upper subscapular, thoracodorsal, lower subscapular, axillary and radial nerves.

Terminal branches:

The five major terminal branches are:

Musculocutaneous	C5-C7	Anterior compartment of arm
Axillary	C5-C6	Deltoid and teres major
Radial	C5-T_1	Extensor comp. of arm & forearm
Median	C5-T_1	Most forearm flexors, thenar muscles
Ulnar	C5-T_1	Most intrinsic hand muscles and medial forearm muscles



Root values are the clinical teaching values; individual contribution may vary.

Relations of the brachial plexus

In the neck

- Roots emerge between the anterior and middle scalene muscles.
- Trunks pass through the posterior triangle of the neck
- The plexus lies superior and posterior to subclavian artery in this region.

Behind the clavicle

The divisions lie behind the clavicle in the costoclavicular region.

In the axilla

- Cords are arranged around the second part of the axillary artery.

- The lateral cord lies lateral to the artery.
- The medial cord lies medial to the artery.
- The posterior cord lies posterior to the artery.

Functions

1. Motor supply to the muscles of the upper limb
2. Sensory supply to most of the upper limb
3. Contribution to autonomic innervation of blood vessels, sweat glands
4. Coordination of movements of the shoulder, arm, forearm and hand.

Clinical anatomy

1. Erb-Duchenne palsy

Usually involves the upper trunk, predominantly C5-C6.

Causes include excessive separation of the head and shoulder, such as during certain birth injuries or trauma.

Typical features:

- Weakness of shoulder abduction and lateral rotation.
- Weakness of elbow flexion and forearm supination.
- characteristic waiter's tip posture in classic presentations.

2. Klumpke palsy

Classically involves the lower trunk, predominantly C8-T1.

It may produce:

- Weakness of intrinsic hand muscles.
- Clawing of the fingers.
- Sensory loss along the medial forearm and hand.

Horner syndrome can occur when sympathetic fibres are also affected

3. Long thoracic nerve injury

Damage to the long thoracic nerve weakens the serratus anterior muscle, resulting in winging of the scapula and difficulty abducting the arm above the head.

Conclusion

The brachial plexus provides the principal nerve supply to the upper limb. Knowledge of its formation, branches and root values is essential for understanding peripheral nerve injuries and clinical examination.

Fascial Spaces of the Hand

1. Introduction

⇒ The hand contains several potential spaces formed by the arrangement of its fascia, muscles, tendons and fibrous septa.

⇒ These spaces are normally collapsed but may become filled with pus or other fluid during infection.

⇒ The fascial spaces are clinically important because infection can spread through them and may lead to serious complications if not treated promptly.

2. Definition

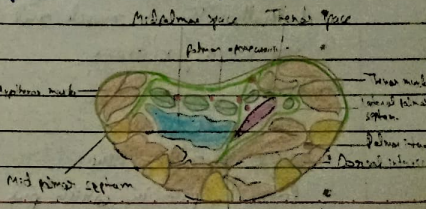
Fascial spaces of the hand are potential spaces formed between the fascial layers, muscles and tendons of the hand, which may become occupied by pus during infection.

3. Classification

The fascial spaces of the hand are classified into:-

A. Palmar space.

1. Thenar space
2. Midpalmar space



B. Dorsal spaces

1. Dorsal subcutaneous space
2. Dorsal subaponeurotic space

Other clinically important spaces and compartments include the pulp spaces of fingers, web spaces and synovial tendon sheaths

4. Palmar Fascial spaces

The palmar spaces are situated in the palm and are separated by fascial septa.

A. Thenar Space

Location: The thenar space is situated in the lateral part of the palm, deep to the flexor tendons of the index finger and related structures.

Boundaries:

- Anteriorly: Flexor tendons of the index finger and first lumbrical muscle
- Posteriorly: Adductor pollicis and first dorsal interosseous muscle.
- Medially: Septum separating it from the midpalmar space.
- Laterally: Thenar muscle and fascial structure of the thumb

Contents:

- Loose areolar tissue
- Potential space in which pus may accumulate.

Communications:

- Distally, it may communicate with first web space.
- Proximally, infections may extend towards the carpal tunnel/forearm.

Clinical Importance: Infection of the thenar space produces swelling and tenderness over the thenar region. The thumb movements may become painful and restricted.

B. Midpalmar Space

Location: The midpalmar space is situated in the central part of the palm, deep to the flexor tendons of the middle, ring and little fingers.

Boundaries:

- Anteriorly: Flexor tendons of the middle, ring and little fingers, their lumbrical muscles and associated fascial structures
- Posteriorly: Palmar interossei and fascia covering them
- Laterally: Septum separating it from the thenar space
- Medially: Medial palmar septum and hypothenar structures

Contents:

- Loose areolar tissue.
- Potential space for accumulation of pus.

Communications:

- Distally, it may communicate with third and 4th web spaces through lumbrical canals.
- Proximally, infection may extend through the carpal tunnel into the forearm.

Clinical importance: Infection of the midpalmar space causes swelling in the central palm, severe pain and restriction of finger movements.

The palmar concavity may become lost due to swelling.

5. Dorsal Fascial Spaces

A. Dorsal Subcutaneous Space

Location: Situated between the skin and the dorsal deep fascia of the hand.

Boundaries:

- Superficially: Skin
- Deeply: Dorsal deep fascia

Contents:

- loose connective tissue
- Superficial veins
- Cutaneous nerves

Clinical importance: Infection may cause marked swelling over the dorsum of the hand because the loose subcutaneous tissue allows accumulation of fluid.

B. Dorsal Subaponeurotic space

Location: Situation deep to the extensor tendons and dorsal aponeurosis, superficial to the dorsal interossei and metacarpal bones.

Boundaries:

- Superficially: Extensor tendons and dorsal aponeurosis
- Deeply: Dorsal interosseous muscle and metacarpal bones

Contents: Loose areolar tissue.

Clinical importance: Infection may produce swelling deep to the extensor tendons and may spread between adjacent tissue planes.

6. Clinical Importance

The fascial spaces of the hand are important in following conditions:

1. Felon: Infection of the pulp space of the finger tip.
2. Tenosynovitis: Infection of a flexor tendon sheath.
3. Fascial space infection: Produces swelling and pain in the thenar region.

4. Midpalmar space infection: Produces swelling and pain in the central palm.

5. Horseshoe abscess: Communication of infection between the radial and ulnar bursae.

6. Spread to forearm: Infection may extend through the carpal tunnel into the deep spaces of the forearm.

Early diagnosis and appropriate treatment are essential to prevent damage to tendons, nerves, blood vessels and muscles.

Conclusion:

⇒ The fascial spaces of the hand are potential spaces formed by fascial and muscular arrangements.

⇒ They are clinically significant because infections can spread through these spaces and may cause severe pain, swelling, restricted movements and functional impairment of the hand.

⇒ Knowledge of their location, boundaries and communications is essential for understanding the spread of hand infections.

Cell Surface Modifications

1. Introduction

The cell surface is modified in different ways to perform specialized functions such as absorption, movement, protection and intercellular communication.

The major cell surface modifications are:

1. Microvilli
2. Cilia
3. Stereocilia
4. Basal infoldings

2. Difference Between cell surface Modification

Feature	Microvilli	Cilia	Stereocilia	Basal infoldings
Definition	Finger-like projection of cell membrane	Hair-like projection from cell surface	Long, non-motile projections.	Infoldings of basal cell membrane
Cytoskeleton	Actin filaments	Microtubules	Actin filaments	Plasma membrane infolding with mitochondria.
Core structure	Actin	Microtubules	Actin	Plasma membrane
Motility	Non-motile	Usually motile some are sensory	Non-motile	Non-motile
Arrangement	closely packed	Usually scattered or organized	Long and irregular	Basal surface.
Main function	Increase surface area for absorption	Movement of substances over cell surface	Absorption and sensory functions	Transport of ions and water.
Examples	Intestinal epithelium, proximal convoluted tubule	Respiratory tract, uterine tube	Epididymis, inner ear	Renal tubules, ducts of salivary glands

3. Cilia: Structure and 9+2 Arrangement

Definition: Cilia are hair-like projections arising from the apical surface of cells. They contain microtubules and help in movement of substances or sensory functions.

Structure of cilia

- Each cilium is covered by plasma membrane.
- The central core is called the axoneme.
- The axoneme contains nine peripheral microtubule doublets and two central microtubules.
- The cilium arises from a basal body containing nine microtubule triplets.
- Dynein arms help in ciliary movement by using ATP.

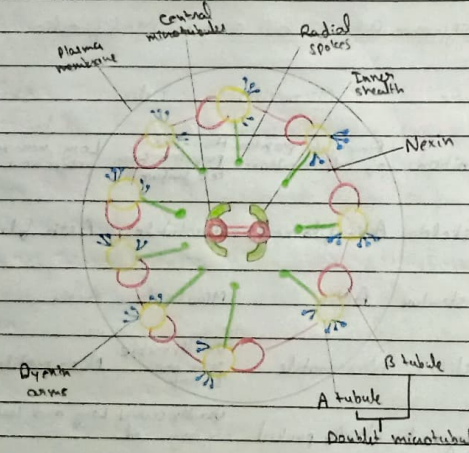


Fig 9+2 arrangement of cilia

Component	Description
9 peripheral doublets	Nine pairs of microtubules arranged in a circle
2 central microtubules	Two single microtubules present in the centre
Dynein arms	ATP-dependent motor proteins responsible for movement
Radial spokes	Connect peripheral doublets to central apparatus
Nexin links	Connect adjacent peripheral doublets
plasma membrane	Cover the axoneme

Mnemonic: Cilia = 9+2, while basal body = 9 triplets + 0 central = (9+0) arrangement.

Functions of cilia

1. Move mucus and trapped particles in respiratory tract.
2. Transport ovum through uterine tube.
3. Help in sensory functions in specialized cells.

Examples

- Ciliated epithelium of trachea and bronchi.
- Ciliated cells of uterine tube.
- Ependymal cells lining ventricles of brain.

4. Microvilli

Definition: Microvilli are small finger-like projections of the plasma membrane that increases the surface area of the cell for absorption.

- Contain actin filaments
- Covered by plasma membrane
- Non-motile.
- Form brush border in intestine and kidney.

Functions: Absorption and increase in surface Area.
Examples: Intestinal epithelia cells and proxial convoluted tubules of kidney

5. Stereocillia

Definition: Stereocillia are long, non-motile projections of cell surface containing actin filaments.

- Longer than microvilli
- Lack the 9+2 microtubules arrangement
- Increase surface area on act as sensory structures.

Examples:

- Epididymis: absorption
- Inner ear: hearing and balance

6. Basal infoldings

Definition: Basal infoldings are folds of the plasma membrane at the basal surface of cells.

- Increase surface area for transport
- Associated with numerous mitochondria.
- Help in active transport of ions and water.

Examples: Renal tubules and striated duct of salivary glands.

7. Conclusion

Cell surface modifications are specialized structures that help cells perform their functions efficiently. Microvilli increase absorption, cilia provide movement, stereocilia perform absorption or sensory functions, and basal infoldings facilitate active transport